

# Biology

Zoology  
(Animals)

Botany  
(Plants)

Microbiology  
(Microscopic organisms)

## Various Branches

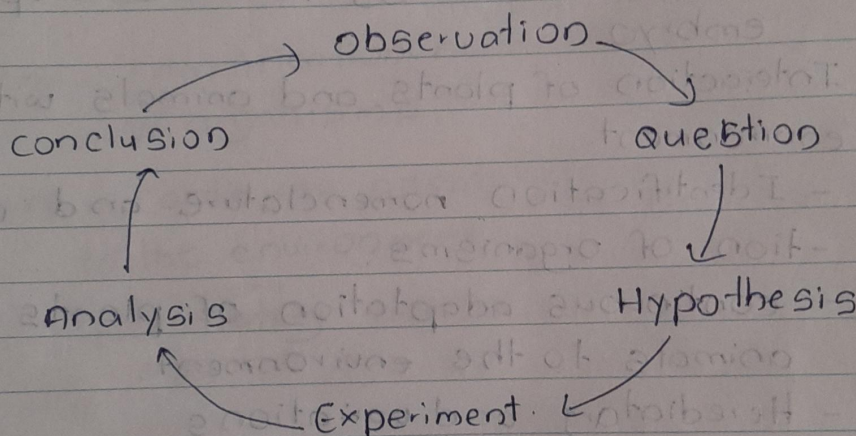
- ① Physiology - organ systems.
- ② Embryology - Formation of gametes, Events from fertilization of egg to the development of embryo.
- ③ Ecology - Interaction of plants and animals with the environment.
- ④ Taxonomy - Identification nomenclature and classification of organisms.
- ⑤ Evolution - continuous adaptation of plants and animals to the environment.
- ⑥ Genetics - Hereditary and variations.



## Biology Relationship.

- ① cell membrane - structure of lipids and proteins - (CHE)
- ② Release of energy during respiration electron transport chain - (CHE)
- ③ Effect of light on flowering. Absorption of different wave length of light - (PHY)

## Scientific Method in Biology.



— observation (start)

- structure
- process
- behavioral pattern etc.



No: .....  
Date: .....

All Living organisms share 5 basic characteristics.

① Order

- All organisms are composed of one or more cells in a highly ordered structure.
- atoms  $\rightarrow$  molecules  $\rightarrow$  cellular organelles  $\rightarrow$  cells.

② Sensitivity

- Living things respond to stimuli
- ex: plants grow toward light.

③ Growth, Reproduction, development.

④ homeostasis.

⑤ Evolution.

- with the time ~~the~~ living things change according to the environment.



# Cell

- \* Cell is the basic unit of structure and function of living things.

cells → Tissues → organs → organ system → Individual.

## cell theory

- \* proposed - schlieden and schwan.

- \* Includes -

① All organisms are composed of one or more cells and metabolism and hereditary occur within these cells.

② cells are the smallest living things, the basic units of organization.

③ cells arise only by division of previously existing cell

## two types of cells

① prokaryotic cell

② Eukaryotic cell



## prokaryotic cell

- \* without true nucleus.
- \* DNA is not enclosed.
- \* no membrane bounded internal structures.

### Structure of prokaryotic cell:

#### \* Bacterial cell (Image)

- cell wall — peptidoglycan
- capsule
- \* cell wall is covered by (slim layer)
- \* additional protection → unfavorable conditions.
- \* Helps to stick to each other. — colonies
- \* Helps to stick to the surface of the food supply or to the host cell.
- plasma membrane
- \* Below the cell membrane
- \* selectively permeable  
(It permits some molecules to pass through it But not others)
- cytoplasm
- \* plasma membrane encloses.  
(It looks granular due to the presence of ribosome)
- \* found — enzymes, carbohydrate, lipid, and genetic material



• Flagella.

- \* many bacteria,  
one or more flagella.

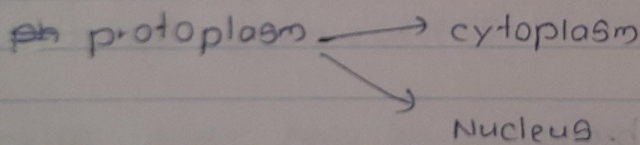
helps - move and to respond to stimuli

week 04

### Eukaryotic cell

- \* highly evolved cells possessing
- \* specialized organelles.
- \* definite membrane
- \* bounded nucleus.
- \* highly organized cytoplasm

Jelly like translucent protoplasm bound by a plasma membrane.



cytoplasm has number of compartments (organelles)  
(carry out different function)

membrane - Mitochondria, Ribosome, Golgi apparatus,  
lysosome.



## Two types Eukaryotic cell.

- ① plant cell
- ② animal cell

major differences

| plant cell                | animal cell              |
|---------------------------|--------------------------|
| i) cell wall              | does not                 |
| ii) large central vacuole | number of small vacuoles |
| iii) chloroplast          | does not.                |

Images plant cell and animal cell

- plasma membrane.
- \* Most favored model - fluid ~~mos~~ mosaic model
- \* proposed - Singer and Nicholson.
- \* made up - lipid, protein and carbohydrates.

Lipids form a fluid bilayer.

head - directed outward

(lipid molecules) polar

hydrophilic.

Tail - directed inward.

non polar

hydrophobic.

Image plasma membrane.



proteins are two types.

## 1. Intrinsic proteins / Integral proteins.

- \* embedded in the lipid bilayer.
- \* held by hydrophobic and hydrophilic parts of the lipid molecules.
- \* cannot be easily dislodged
- \* Some proteins molecules are small - part project at the surface.

Large - projects at both surfaces.

## 2. Extrinsic proteins / peripheral proteins.

- \* superficially attached to the lipid layer.
- \* can be easily removed.

carbohydrate - antennae projecting on the surface

(The cells recognize one another by these)



- Cell wall.

- primary cell wall

on the middle lamella (within the cell)

consists - cellulose microfibrils running through the matrix

- Secondary cell wall.

built up by addition of cellulose layers on the inside surface of primary cell wall.

- plasmodesmata.

interrupted by minute pores called plasmodesmata

- Image -



## ① chloroplast

- Surrounded by 2 membranes
- Separated by a space.
- these membranes enclosed a fluid space called stroma / matrix



membrane system is suspended.

### thylakoids / Lamella.

- fluid filled sacs.  
(Many flattened)
- grana
- thylakoids stack to  
integrated  
integrated lamella.
- various grana are jointed by

### Image chloroplast

## 2) Mitochondria.

- power-house of the cell.
- specialize - synthesize of ATP  
The usable chemical energy.
- sausage shaped.
- bound by 2 membrane
- separated by a space.



- Inner membrane - folds into numerous folds called cristae.  
cristae is packed with elementary particles/ granules.
- matrix is lying inside the inner membrane.

### 3) Lysosome .

- membrane bound - single.
- digestive vesicle
- contain - <sup>5555</sup> mixture of digestive enzymes.

- Image -

### 4) Ribosome

- Spherical bodies found in the cytoplasm of the cell.
- composed - large and small subunit.
- They serve as sites of proteins synthesis.

- Image -

### 5) Nucleus

- largest and most easily seen organelle.
- core of the cell which command all metabolic activities
- In animal cell, nucleus is located in the central region
- Large amount of DNA is packed in the nucleus
- Roughly spherical in shape.



- Encased by 2 membranes!
- Nucleolus → is the site within the nucleus where ribosome are made.
- It contains large amount of DNA and RNA
- nucleolus is filled by the transparent granular nucleoplasm.

### 3) Mitochondrion

- bean-shaped organelle.
- contains digestive vesicles.
- contains a mixture of digestive enzymes.

- Image -

### 4) Ribosome

- spherical bodies found in the cytoplasm of the cell.
- composed of large and small subunits.
- they serve as sites of protein synthesis.

- Image -

### 5) Nucleus

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# week 6

## classification.

No. ....

Date: .....

Two kingdom. → plants and animal

\* Linnaeus, divided to group 2 with.

Six kingdom, only all most.

1. Archaeobacteria (bacteria)
2. Eubacteria (bacteria and cyanobacteria/blue green algae)
3. protista (algae and slime moulds)
4. Fungi (Mushroom, yeasts, moulds)
5. plantae (plants)
6. Animalia (animal)

viruses → addition.

- Sub microscopic agents
- Seen only through the electron microscope.
- unable to grow and reproduce outside living cells.
- Share both living and non living characteristics of viruses.

### Living

1. possess genetic material DNA/RNA
2. possess protein coat
3. can replicate (inside a living cell)

### Non living

1. non cellular
2. cannot reproduce by themselves.
3. Don't possess ribosomes, mitochondria etc.
4. Only have one type of nucleic acid.



## Kingdom I - Archaeobacteria.

- diverge group of bacteria.
- Archaeobacteria different → Eubacteria.
  - from the places where they live.

live → extreme conditions which are not suitable for other normal life forms.

3 major groups.

1. methanogens
2. Halophiles
3. Thermophiles.

### 1. Methanogens.

live → oxygen less environments.

found in → sewage treatment plants, bogs, intestines of some animals.

### 2. Halophiles

thrive in high salt concentrations.  
(sea water)

### 3. Thermophiles

heat-loving bacteria  
(hot water springs)



## Kingdom 2 - Eubacteria.

Includes - Bacteria and cyanobacteria.

### Bacteria

- Tiny minute organisms
- cannot be seen naked eye.

Image bacteria.

### cyanobacteria.

- prokaryotic
- photosynthetic (can make their own food)

Ex: Microcystis

## kingdom 3 - protista. (Algae and slime moulds)

- All are Eukaryotes
- All algae contain chlorophyll
- depending on the type of algae along with chlorophyll 'a' they can have chlorophyll b, c, d or e
- algae are all photosynthetic protista
- Algae may occur in oceans, lakes, tanks, rivers, streams, damp soil, bodies of other organisms.

Ex: Chlamydomonas, Spirogyra.



## Kingdom - 4 fungi

- non-photosynthetic (no-chlorophyll)
- Eukaryotic organisms
- heterotrophic (get their food from others)
- Most are small, microscopic, filamentous branched forms that bear spores.

ex: Mushrooms

## kingdom 5 - plantae.

Two types.

1. vascular.
2. non vascular.

non vascular → more primitive

ex: Thalloid liverworts

Mosses

ferns

vascular plants.

two types.

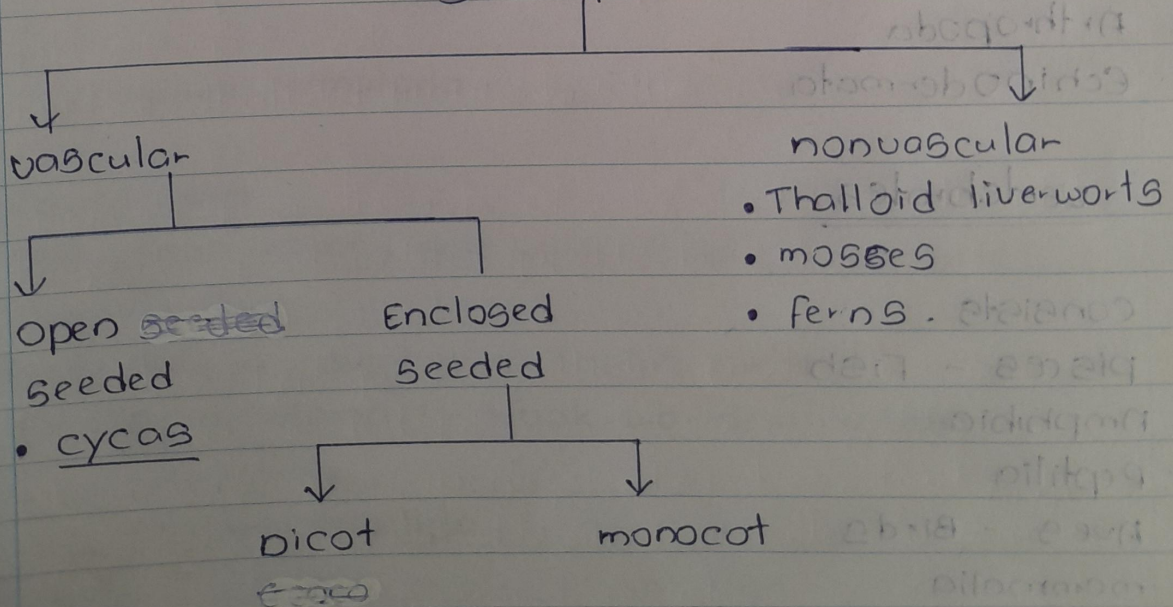
1. open seeded plants. (ex: cycas) Gymnosperms
2. Enclosed seeded plants. Angiosperms
  - Dicot plants
  - monocot plants.



## Differences between dicot and monocot plants.

|                    | Monocot                                            | dicot.                                               |
|--------------------|----------------------------------------------------|------------------------------------------------------|
| Embryos            | one cotyledon                                      | two cotyledon                                        |
| leaf -<br>venation | veins usually parallel                             | veins usually netlike                                |
| stems              | vascular bundles<br>usually complexly<br>arranged. | vascular bundles usually<br>arranged in ring.        |
| roots              | Fibrous root systems                               | Taproot usually present.                             |
| flowers            | Floral parts usually<br>in multiple of three       | Floral parts usually in<br>multiples of four or five |

## kingdom plantae.





## Kingdom 6 - Animalia

consists

- Multi cellular (composed of more than one cell)
- heterotrophic (Feed on other organisms)

two main groups

I Invertebrates.

II vertebrates.

Invertebrates

consists

1. porifera - sponges
2. cnidaria
3. plathelminthes
4. Nematoda
5. Annelida
6. Mollusca
7. Arthropoda
8. Echinodermata

vertebrates

consists

1. pisces - Fish
2. Amphibia
3. Reptilia
4. Aves - Birds
5. mammalia



## phylum - porifera

- called pore bearers as their body wall contain minute pores.
- All are aquatic marine or freshwater.

Ex: Euspongia

## phylum - cnidaria.

- Radial symmetrical animal
- specialized cell - cnidocytes "stinging cells"
- Ex: Jelly fish.

## phylum - plathyhelminthes (Flat worms)

- Soft, dorso - ventrally flattened animals with bilateral symmetry.
- Most are parasitic
- Ex: Taenia solium

## phylum Nematoda.

- Round worm
- Soil, fresh water, salt water and as parasites in the body.
- Body wall is covered by a tough cuticle
- Ex: Ancylostoma - Hook worm infection.

## phylum annelida.

- Elongated segmented body
- Body is covered by a transparent cuticle



## Phylum - Mollusca.

- live - Fresh water, marine water and on land
- soft bodies
- muscular foot used for locomotion.

ex: snail

## Phylum Arthropoda.

- largest group
- Reasons for the success of Arthropods

### 1. chitinous exoskeleton

- chitinous (made up of chitine) cuticle
- light weight and tough
- protect → predators
- It is impervious to water and so protects the body from drying.
- Insects have chitinous wings that help them to fly.
- The biting mouth parts, lenses for the eye and ~~see~~ sensory organs are also made up of chitin.

### 2. Segmentation.

- Segmentation into distinct region as head, thorax and abdomen
- each one is specialized to perform a specific function.



### 3. Jointed appendages.

- The appendages have been modified to perform different functions.
- perform different function.
- increases the efficiency and complexity of the activity

### 4. Ecdysis / molting.

- chitinous skeleton doesn't allow the body to grow and hence to overcome it, ecdysis has evolved.

### phylum Echinodermata

- marine animals
- spiny skinned animals.
- The spines fixed or movable.
- Radial symmetrical animals with water vascular system
- Have an anterior - posterior flattened body.

### phylum chordata.

#### 1. class - pisces (Fishes)

- Two types:
  1. cartilaginous fishes.
  2. Bony fishes

cartilaginous fishes cartilaginous skeleton.

They don't have swim bladder.

They are viviparous - live bearers.



## Bony fishes.

- have bony skeleton
- have spindle gills
- respire through gills
- have a swim bladder - help rise and sink
- have two chambered heart

## 2. class - Amphibia.

- live - water and land
- smooth and moist skin
- two pairs of limbs
- heart is three chambered.

Ex: tree frog

## 3. class - Reptilia.

- dry horny scaly skin
- two pairs of limbs
- heart is three chambered

Ex: python

## 4. class - Aves.

Distinct features of birds is flight.

1. ~~stream~~ body covered by feathers - help to reduce the resistance.
2. fore limbs are modified into wings.
3. light, porous bones with air cavities to reduce the weight.



4. Strong breast bone called keel to which the flight muscles are attached.
5. Heart is four chambered. Which allows complete separation of the oxygenated blood from the deoxygenated blood.
6. Hind limbs are adapted for perching
7. tail used - changing the direction.
8. constant body temperature called warm blooded.
9. Maintain a high metabolic rate in order to lift off the body from the ground.
10. The heart of the birds beats faster in order to supply large amount of oxygenated blood in to the body.
11. large eyes and strong sense of sight  $\rightarrow$  field of vision is also great.
12. A remarkable ability to judge the distance from a close range as well as from a great height.
13. disappearance of right ovary and right oviduct reduces weight and is an adaptation for flight.

### class Mammalia.

- viviparous and nourish - milk secreted from mammary-glands.
- covered with hair
- External ear present
- Spiny and eater is an exceptional that egg laying mammal
- In kangaroo young ones - into the pouch and feed on the milk.
- shrews are the smallest insectivorous mammal
- Bats are flying mammal. Their fore limbs are modified into wings.



- Whales, Dolphins are aquatic mammals. Fore limbs are modified into flippers.
- Man, Monkey, Gorilla, are highly evolved mammals.
- They have opposable thumb. Intelligent animal.

Week 10

Communities and Ecosystems

Community - All the organisms that live together in a place called a community.

co-evolution.

(plants, animals, protists) live together and have changed and adjusted to one another continually over a period of millions of years.

Such interactions are called coevolution.

symbiotic relationship.

Kinds of Symbiotic

1. commensalism

2. Mutualism

3. parasitism



## Commensalism.

No .....  
Date: .....

- gives benefits for one species and neither hurts nor helps the other.

Physically attached to members of another.

Ex: Epiphytes

the plants grow on the branches of another plants.

Host plant is unharmed and the epiphyte benefits by having the substrate for growing.

ex: orchid

## Mutualism.

- both species benefits.

Ex: ants and aphids.

Aphids are small insects that suck fluids from the phloem.

Ex: ants and aphids.

Aphids are small insects that suck fluids from the phloem of living plants with their piercing mouth parts.

They excrete them through their anus.

Ants carry the aphids to new plants and then consume the excreted materials of aphids as food.

So, aphids get new plant and ants get food.



## Parasitism.

Smaller than the prey that closely associated with it.

harmful to the prey organisms and beneficial to the parasite.

Ex: lice mites.

## Dynamics of the Ecosystem.

every ecosystem has regulated transfer of energy and a controlled cycling of nutrients.

All of the substances cycle through ecosystems in biochemical cycle.

1. water cycle
2. carbon cycle
3. Nitrogen cycle.

## Trophic levels

- An ecosystem is composed of autotrophs and heterotrophs.

Autotrophs - plants, algae, some bacteria

- capture light energy and manufacture their own food.

primary producers

Heterotrophs - animals, fungi and most other organisms

obtain organic molecules that have been synthesized by autotrophs.

\* consumers



different levels of consumers.

Primary consumers - herbivores (Feed directly on the green plant)

Secondary consumers - carnivores and parasites (Feed in and on the herbivores)

Decomposers - break down the organic matter accumulated in the bodies of other organisms.

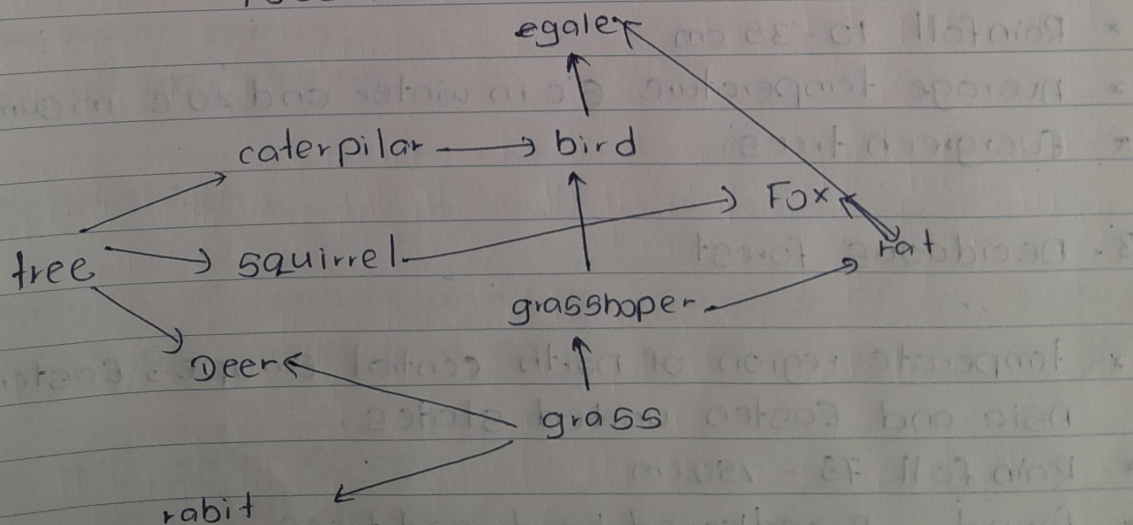
Organisms from each trophic level feed on one another make up a series called "Food chain".

Flower → caterpillar → Frog → Snake → owl  
(producer) (consumer) (consumer) (consumer) (consumer)

When diagrammed it appears as a series of branching lines, than a straight line.

It is called a food chain

Food web.





# Biomes

(14)

- \* largest terrestrial community / largest ecological unit
- \* specific kind of plants and animals
- \* distinct climatic conditions
- \* distributed on earth as broad belts

## The major biomes.

### 1. Tundra

- \* Found - Asia, Europe and North America
- \* Rainfall - below 25 cm
- \* poor light, extremely low temperature.
- \* snowfall is scanty
- \* sparse vegetation
- \* Found - dwarf willow trees, lichens, mosses.

### 2. Taiga.

- \* east west band across, North America, Europe and Asia.
- \* Rainfall 10-35 cm
- \* Average temperature 6°C in winter and 20°C in summer
- \* Evergreen trees.

### 3. Deciduous forest

- \* temperate region of north central Europe, Eastern Asia and Eastern United States.
- \* Rainfall 75 - 150 cm
- \* found - Broadleaved hardwood trees.



#### 4. Tropical rain forest

- \* Occured in - equatorial regions
- \* Rain fall exceeds 140cm
- \* warm and humid climate
- \* Luxuriant and varied vegetation

#### 5. chaparral

- \* extends along Mediterranean sea, pasific coast of South America, chilli, South Africa and South Australia
- \* Limited winter rain followed by drought.
- \* vegetation composed of broad - leaved evergreen shrubs, bushes, and small trees usually less than 2.5m tall, together they often form dense thickets.

#### 6. Tropical savannah

- \* South America, Africa, Asia and Australia
- \* ~~Warm~~ Warm climate.
- \* Rain fall 100 - 150 cm

#### 7. grasslands.

- \* prairies of Canada and USA, pampas of South America  
stepps of Europe and Asia, veldts of Africa.
- \* Rain fall - 25 - 75 cm.
- \* Short and tall grasses found.



## 8. Deserts

- \* South western USA, Mexico, coastal area of Chile and Peru, North Africa Sahara desert -
- \* Extremely low rainfall
- \* Less than 25 cm
- \* high evaporation
- \* found - hardy grasses.